Georgios Petrakis

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Profile: Informatics and Telecommunications graduate with hands-on experience in secure software development. Creator of Efaeto, with research contributions in digital identity, privacy-preserving access control and distributed systems.

Education

University of Ioannina — Arta, Greece
BSc in Informatics and Telecommunications — *Sep 2022 to Jun 2026*

- **Final grade: 7.11/10 (Very Good).** 240 ECTS required; 250 recorded. Completed 26 Jun 2026; awarded 2 Oct 2026.
- **Thesis (Parts I and II: 10/10):** "Design and Implementation of a High-Throughput Order Management Application for Large-Scale Events".
- **Coursework:** IoT, Sensor Networks, Special Topics in Networks and Advanced Topics in Network Programming (10/10); Network Management and Statistical Machine Learning (9/10); Systems Security (7/10).

Certificate of Pedagogical and Teaching Competence, University of Ioannina, issued 2 October 2026. Planned lessons and taught GLOSSA programming to Grade 11 pupils in Arta; **teaching practicum: 9.8/10**.
Computer and Network Technician, Level 4, 5th Vocational High School of Heraklion, Sep 2019 to Jun 2022. **Grade: 18.8/20**.

Experience

Cybersecurity Engineering Intern — Mar to Apr 2026
KIC / EPIC Laboratory, University of Ioannina — *Arta, Greece*

- Developed the full web application and its security: React frontend, Node.js/Express backend, MongoDB storage and API integration for Roblox authentication, age verification and access control.
- Implemented digital credentials and single-use tokens using the eIDAS issuer/holder/verifier model, with rate limiting and protection against NoSQL injection and XSS.
- Contributed to research writing on security architecture and the protection of minors in virtual environments.

Efaeto — Current project
Creator and Software Developer — *Heraklion, Greece*

- Develop a local-first hospitality platform with real-time orders and inventory, an IndexedDB outbox for offline work, and recoverable stock journals.
- Built cloud licensing and administration, encrypted backups, and Ed25519-signed updates with integrity checks and automatic rollback.

Research and Projects

Local-First Event Order Management System | *Bachelor's Thesis*

- Built an event-driven MERN system for order management at large events with role-based access control, Socket.IO synchronisation and ESC/POS printing over raw TCP with Greek character support.
- Deployed for **seven operational days** at a local cultural festival, recording **694 transactions**.
- Tested with Apache JMeter on localhost: **178 ms mean response time** in an extended run with **200 simulated clients**.

Arduino WSN ESPNow: Built an ESP32 sensor network in C++ using ESP-NOW, GPIO control, hardware interrupts and state-machine logic.
Social Network Platform: Built a React, Node.js, Express and MongoDB application with REST APIs, persistent storage, authentication and authorisation.
Conference paper (presented): "A Local-First Event-Driven Order Management System for Temporary Events", **IEEE SEEDA-CECNSM 2026**, Aristotle University of Thessaloniki, 18 to 20 September 2026. Co-authored with Spiridoula V. Margariti and Christos Gogos.
Research manuscript: "Bridging the Age Enforcement Gap in Metaverse Platforms: A Privacy-Preserving Age-Aware Access Control Architecture for Child Protection". Co-author; developed the web application and implemented its security controls. Manuscript in preparation for submission to *Computers & Security*.

Technical Skills

Programming: JavaScript, TypeScript, Python, C/C++, SQL, Lua, VHDL, PHP
Web and data: React, Node.js, Express, Socket.IO/WebSockets, Tailwind CSS, MongoDB, SQLite
Tools: Git, Docker, Linux, Wireshark, Cisco Packet Tracer, Roblox Studio, Arduino IDE

Languages and Qualifications

Greek: Native. **English:** C2, LRN Level 3 Certificate in ESOL International (CEF C2), issued 18 September 2018.
Develop AI-Powered Prototypes in Google AI Studio, Google Cloud Skills Boost, May 2026.
Introduction to Cybersecurity; NDG Linux Unhatched, Cisco Networking Academy, February 2023.
State Certificate of Informatics, Greek Ministry of Education, July 2022.
Basic Life Support (BLS), European Resuscitation Council, November 2024.
Online Student Festival of Digital Creation, Regional Educational Planning Centre of Crete, April 2021. Workshops in computing, virtual reality, microcontrollers and digital design.